Dayananda Sagar University

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Department of Computer Science & Technology

Project Phase -I Report

ASPIRE360: AI-POWERED CAREER PATHWAYS AND INTERACTIVE SIMULATIONS

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CERTIFICATE

This is to certify that the work titled **"ASPIRE360: AI-POWERED CAREER PATHWAYS AND INTERACTIVE SIMULATIONS "** is carried out by Khushi Shah - ENG21CT0015, Niharika R - ENG21CT0028, Sohana R - ENG21CT0038 and Tanya Singh - ENG21CT0044 Bonafide students of Bachelor of Technology in Computer Science and Technology at the School of Engineering, Dayananda Sagar University, Bangalore in partial fulfillment for the award of degree in Bachelor of Technology in Computer Science and Technology, during the year **2024-2025**.

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DECLARATION

We, **Khushi Shah (ENG21CT0015), Niharika R (ENG21CT0028), Sohana R (ENG21CT0038) and Tanya Singh (ENG21CT0044)**, are students of the seventh semester B.Tech in Computer Science and Technology, at School of Engineering, Dayananda Sagar University, hereby declare that the project phase - I titled "**ASPIRE360: AI-POWERED CAREER PATHWAYS AND INTERACTIVE SIMULATIONS**" has been carried out by us and submitted in partial fulfillment for the award of degree in Bachelor of Technology in Computer Science and Technology during the academic year 2023-2024.

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Place: Bangalore
Date:

ABSTRACT

This project envisions the creation of an innovative, **AI-powered career counseling** platform designed to empower students in making well-informed and personalized career decisions. Harnessing the transformative capabilities of artificial intelligence, the platform meticulously analyzes a range of key factors—including students' interests, inherent strengths, and preferred work environments—to provide tailored career recommendations that resonate with each individual's unique profile. By aligning students' innate potential with their professional aspirations, the platform fosters a deeper sense of clarity and purpose in their career journeys. In addition to its AI-driven personalization, the platform incorporates cutting-edge **Augmented Reality (AR)** technology to revolutionize the way students explore career options. Through immersive and interactive AR simulations, students can step into realistic virtual environments that mirror various professional settings. These simulations offer an unparalleled opportunity to experience the day-to-day realities of different career paths, enabling students to make choices that are not only informed by data but also grounded in experiential insights. This innovative fusion of **AI and AR bridges** the gap between theoretical understanding and practical exposure, fostering holistic career exploration in a highly engaging and interactive manner. By addressing the critical challenges of limited career awareness and misalignment between aspirations and opportunities, this platform aims to illuminate the vast spectrum of career possibilities available to students.

Keywords:

AI-driven career counseling, Augmented Reality (AR), holistic career decision-making, career path alignment, skill gap reduction, empowering student aspirations, AI and AR integration, data-driven career insights, student empowerment.

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INTRODUCTION

The challenge of students making poor career choices is a pervasive issue that has far-reaching implications for individuals and society alike. Research shows that an astounding 90% of students end up pursuing careers that do not align with their natural skills, interests, or passions. This misalignment is often the result of inadequate career guidance during their formative years. When students make career decisions without proper understanding or support, they frequently experience dissatisfaction, underperformance, and, in severe cases, mental health challenges such as stress, anxiety, and depression. A deeper dive into the issue reveals the root causes of this problem. A recent survey found that 93% of students are familiar with only a limited set of career options—primarily engineering, medicine, law, and information technology. This lack of awareness stems from limited exposure to the vast array of professional opportunities available. In reality, there are over 800 distinct career paths that students could explore, yet many remain undiscovered due to the constraints of societal expectations, peer pressure, and outdated norms. As a result, students often feel compelled to follow predetermined paths rather than pursuing careers that genuinely resonate with their unique talents and aspirations. The consequences of these uninformed decisions are significant. For individuals, it manifests as widespread job dissatisfaction and, often, a sense of being stuck in roles that do not fulfill them. On a broader scale, the workforce suffers as well. Approximately 75% of organizations report a significant skills gap, indicating a mismatch between the competencies employees possess and the demands of the job market. This disconnect highlights a systemic issue that perpetuates unemployment and underemployment, even among educated individuals. Despite the critical nature of this issue, the availability and quality of career counseling remain insufficient in most educational institutions. Current systems are often unable to provide personalized, comprehensive guidance that addresses students' individual interests, strengths, and future aspirations. Experts emphasize the urgent need for a structured, well-rounded approach to career counseling.

LITERATURE SURVEY

1.	Envisioning Tomorrow: AI Powered Career Counselling	Madhuri Ghuge	2023	The paper introduces provides personalized career recommendations to students. It improves upon traditional counselling methods.	The authors use artificial intelligence, with the Adaboost classifier showing the best performance. The methodology combines K-Nearest Neighbor for skill categorization and K-Means Clustering for department choices to provide personalized career recommendations.	Future enhancements include adding psychometric tests, optimizing algorithms, increasing user engagement, and incorporating industry feedback to create a more dynamic career counselling system.
2.	AI-Powered Academic Guidance and Counseling System Based on Student Profile and Interests	Hajar Majjate	2024	This research presents a novel educational counselling system designed to support high school seniors during the college application process. The system aims to help students select suitable colleges or universities based on their academic background and institutional requirements. It provides valuable insights into students' chances of acceptance at preferred institutions while also suggesting alternative academic opportunities at similar universities.	The proposed methodology involved implementing a user-friendly interface using PyCharm software and evaluating various different machine learning algorithms for prediction accuracy. After evaluation, the Huber Regressor emerged as the most accurate model. The system incorporates both content-based and popularity-based recommendation approaches - the content-based approach suggests similar universities based on university descriptions, while the popularity-based approach shows trending university choices on the homepage.	The authors plan to expand the database to include more national and international institutions, incorporate additional student experiences, and improve the web interface to make it more informative and user-friendly. The goal is to create a more comprehensive and robust educational guidance resource.
3.	AI based SSS for Educations System	Saddam H. M. Issa	2024	The paper explores the potential of AI technologies to enhance academic advising and counselling services in higher education. The authors argue that AI-enabled systems can improve the precision and accessibility of academic advice, streamline counselling processes, and offer customized support for diverse student needs. The paper reviews current implementations and analyses their impact, suggesting that AI integration can significantly improve the academic experience.	The authors proposed a systematic implementation of AI technologies through a multi-step process: requirements analysis, data collection/preparation, machine learning model development (using algorithms like decision trees, random forests, KNN, SVM), NLP integration for automated responses, and system integration with continuous monitoring.	The system could be enhanced by incorporating deep learning, expanding data sources, improving emotional intelligence capabilities, integrating VR/AR technologies, and developing better adaptation to educational and job market trends. These improvements would focus on creating more personalized and effective student support services.
4.	Artificial Intelligence for Career Guidance – Current Requirements and Prospects for the Future	Stina Westman	2021	This paper explores the potential role and implications of artificial intelligence (AI) in career guidance services within higher education. The authors examine the possibilities, requirements, and ethical considerations for integrating AI into career guidance processes.	The authors employed a multiple methods approach, combining focus groups, scenario workshops, and practical trials. These trials utilized machine learning, data mining techniques, and AI components integrated with existing commercial mobile student services.	The authors recommend addressing ethical issues in AI-enhanced guidance services, and further exploring and refining the concept of maturity levels for AI involvement in guidance processes.
5.	D-Learning: An Experimental Approach to Determining Student Learning Outcomes Using Augmented Reality (AR) Technology	Jadranko Tuta	2024	The research evaluates the impact of active learning using augmented reality (AR) on students' knowledge outcomes. It compares different teaching approaches and demonstrates that cadets exposed to all forms of AR instruction achieved significantly higher knowledge scores. The findings support the potential of AR in enhancing educational methods.	The authors proposed a methodology involving three active learning frameworks: Situated, Games, and Research-based learning, utilizing the augmented reality application HaubicAR. The research assessed perceptions and knowledge through surveys and tests, with statistical analyses for validation.	Future enhancements suggest applying this methodology across various study programs to further explore AR's educational potential.

PROJECT SPECIFICATIONS

Technical Specifications

Backend:

Language: Python (Django/Flask)
Database: PostgreSQL for structured data and MongoDB for unstructured data.
AI Frameworks: TensorFlow, PyTorch, or Scikit-learn for machine learning models.
APIs: RESTful APIs for seamless communication between services.

Frontend:

Framework: React.js or Angular for an interactive UI.
Libraries: Three.js or Unity for 3D career simulations.
Design: Responsive and accessible design using Bootstrap or Material UI.
Infrastructure:

Cloud Services:
AWS or Google Cloud for scalable hosting and data processing.
Storage: S3 for media assets and simulations, RDS for relational data.
Security: OAuth 2.0 for authentication, SSL for secure data transmission.
Target Audience
Primary Users: Students aged 13–25 exploring career options.
Secondary Users: Educators, career counselors, and parents.
Institutional Users: Schools, colleges, and career counseling centers.

Development Timeline

Phase 1: Research and Planning
Define user personas and requirements.
Gather and curate career database content.

Phase 2: Prototype Development
Build AI models for career matching.
Develop basic UI/UX and simulations.

Phase 3: Platform Development
Implement core features and backend integration.
Test and refine simulations and AI recommendations.

Phase 4: Beta Testing
Conduct user testing with students, educators, and institutions.
Collect feedback for improvements.

Phase 5: Launch and Deployment
Deploy the platform to schools and career centers.
Offer onboarding and training for educators.

KEY FEATURES:

1. AI-Powered Career Matching:

AI algorithms analyze student profiles, including interests, academic performance, and personality traits.
Machine learning models predict suitable career paths.

2. Virtual Career Simulations:

Interactive, 3D simulations for professions such as engineering, medicine, law, arts, IT, and more.
Scenario-based tasks to give students a hands-on feel for different careers.

3. Dynamic Career Database:

Detailed descriptions of career paths, required qualifications, job outlooks, salary ranges, and growth opportunities.
Integration of labor market data for real-time insights.

PROBLEM STATEMENT

Inadequate career counseling within schools remains a significant challenge, leading many students to make poor career choices that can have long-term repercussions. The absence of effective guidance often leaves students ill-prepared to navigate the complex world of career opportunities, resulting in widespread job dissatisfaction, mismatched skills, and rising unemployment rates. This issue is particularly pronounced in regions such as India, where the lack of robust career counseling systems exacerbates the problem. Several factors contribute to this concerning trend. Students often have limited exposure to the vast spectrum of career possibilities, with their choices typically confined to conventional fields like engineering, medicine, law, or IT. This narrow perspective is further compounded by societal pressures, including expectations from family and peers, which often push students toward predetermined paths. Without access to personalized guidance that aligns their decisions with their innate skills, passions, and aspirations, students are left to make choices that do not truly reflect their potential. The implications of this systemic issue are far-reaching. When students enter careers misaligned with their strengths or interests, they are more likely to experience dissatisfaction and disengagement. Over time, this misalignment contributes to underperformance in professional roles, perpetuating a cycle of unfulfilled potential and skill shortages in key industries. Addressing this multifaceted challenge requires a transformative approach that moves beyond traditional career counseling methods. There is an urgent need for a dynamic, technology-driven solution capable of empowering students to explore diverse career options tailored to their unique profiles. By leveraging advancements in artificial intelligence, data analytics, and interactive tools, such a system can provide personalized career guidance, enabling students to make informed decisions that align with their individual abilities and interests. This innovative approach would not only expand students' awareness of the myriad career opportunities available but also equip them with the confidence and clarity to pursue fulfilling and sustainable paths.

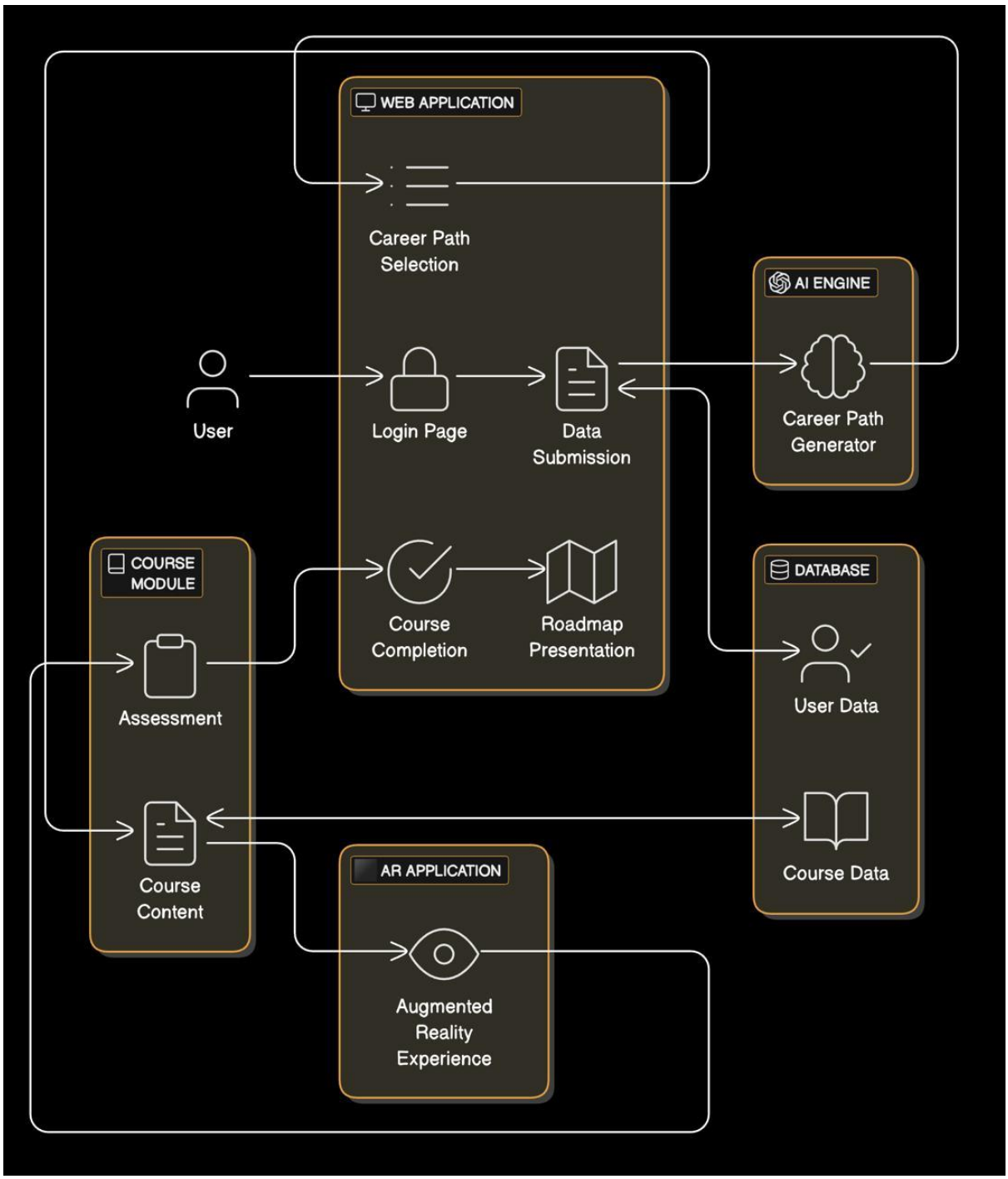
RELEVANCE OF THE PROBLEM

1. The issue of inadequate career counseling holds significant relevance on both individual and societal levels. On a personal level, the absence of proper guidance often leads students to make uninformed career choices, placing them in roles that fail to align with their skills, interests, or passions. This misalignment not only results in job dissatisfaction but also impacts mental well-being, leading to stress, anxiety, and a pervasive sense of unfulfillment. Addressing this problem is essential to empower individuals to realize their full potential, fostering personal growth and enhancing their overall quality of life.

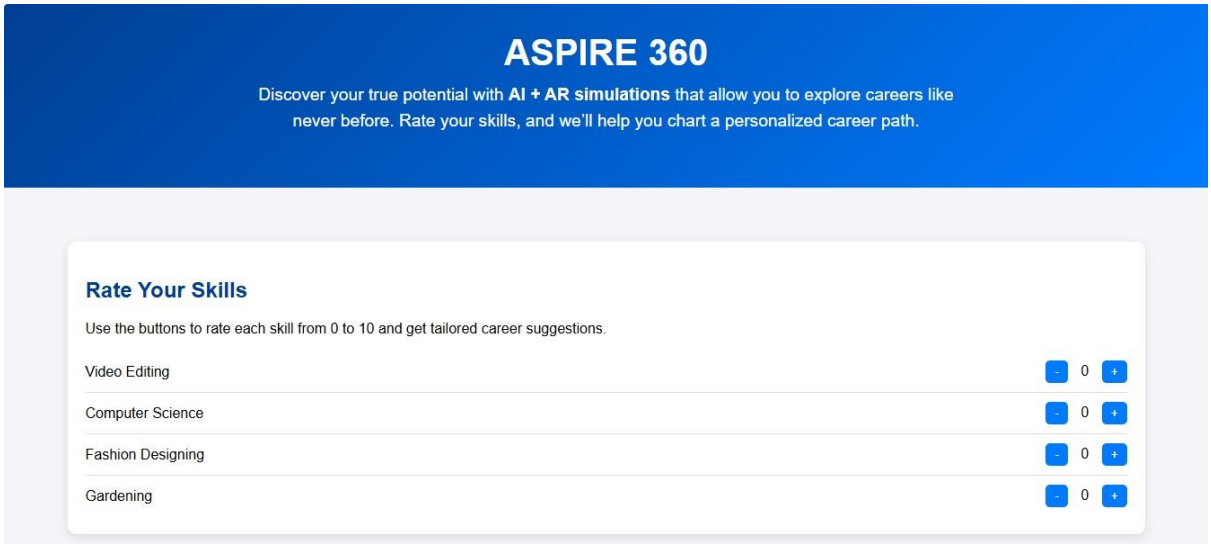
2. On a broader scale, the consequences of poor career guidance ripple through the economy and workforce. Mismatched career choices contribute to a significant skills gap, leaving industries struggling to find professionals with the required competencies. This misalignment hinders organizational productivity and innovation, while also perpetuating unemployment and underemployment. By addressing this issue through structured and effective career counseling, we can align individual capabilities with market demands, creating a more efficient, satisfied workforce and driving sustainable economic growth.

3. Impact on Education Systems and Student Engagement:
The absence of adequate career counseling also undermines the effectiveness of education systems. Without a clear understanding of how their studies connect to future career opportunities, students often lack motivation and engagement in their academic pursuits. This disconnection can lead to higher dropout rates, lower academic performance, and a diminished sense of purpose in education. By integrating career counseling into the education system, students can better appreciate the relevance of their studies to real-world applications, fostering greater enthusiasm, commitment, and long-term success in their educational and professional journeys.

SYSTEM ARCHITECTURE



IMPLEMENTATION/POC



The significance of ASPIRE360 lies in its ability to address a critical gap in career guidance by leveraging technology to empower students in making informed decisions about their futures. Traditional career counseling methods often fail to provide personalized and comprehensive guidance, leaving many students unaware of the diverse career opportunities available or misaligned with roles that suit their skills and passions. ASPIRE360 introduces a dynamic, AI-powered platform that not only expands awareness of career possibilities but also tailors recommendations to individual aptitudes and interests. By incorporating interactive tools, such as skill assessments and career simulations, the platform engages students in a meaningful exploration of their potential paths, fostering self-awareness and confidence in their choices. Additionally, ASPIRE360 contributes to bridging the workforce skills gap by aligning student capabilities with market demands, ensuring they are better equipped to thrive in their chosen professions. This transformative approach not only enhances personal growth and satisfaction but also drives economic development by creating a more skilled, competent, and fulfilled workforce.

CONCLUSION

In conclusion, the issue of inadequate career counselling is a multifaceted challenge that profoundly impacts both individual lives and the broader societal landscape. For students, the absence of effective guidance often leads to misguided career choices, resulting in job dissatisfaction, underperformance, and adverse effects on mental health, such as stress and anxiety. On a larger scale, these uninformed decisions contribute to a significant skills gap in the workforce, where industries struggle to find professionals with the necessary competencies, thereby limiting productivity, innovation, and economic growth. Addressing this pressing issue requires a holistic and forward-thinking approach to career guidance. By leveraging technology-driven solutions, such as AI-powered tools and immersive career simulations, we can offer students the personalized support they need to explore diverse opportunities and make well-informed decisions. Such systems would not only expand awareness of the countless career paths available but also align choices with individual interests, strengths, and aspirations. Ultimately, investing in comprehensive and dynamic career counselling systems is an investment in the future. It enables students to realize their full potential, fostering a generation of confident, skilled, and fulfilled individuals. At the same time, it bridges the gap between education and workforce demands, creating a thriving, adaptable, and sustainable economy.

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